EMORY UNIVERSITY

ROLLINS SCHOOL OF PUBLIC HEALTH

BIOS 509 | APPLIED LINEAR MODELS

Statistical Modeling of Diabetes Outcomes Using SAS

Take-home Project

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Graduate Research Report

INDIVIDUAL CONTRIBUTION

This project was completed collaboratively. Conglin Bao independently drafted Sections 3-5: Methods, Results, and Discussion.

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1. Abstract

Type 2 diabetes mellitus (T2DM) remains a major contributor to vascular disease, with baseline glycaemic status varying by race. Data on 487 adult male T2DM patients (120 White, 135 Black, 107 Pacific Islander, 125 Asian) seen at a large urban clinic were analysed. A one-way fixed-effects ANOVA revealed significant racial differences in fasting glucose ($F = 7.11$, $p = 0.0001$). Reported group contrasts suggested differences in glucose levels, and the combined White and Asian group mean was
9.31 mg/dL below that of Black and Pacific Islanders ($p < 0.0001$).

We next assessed whether race modified the relationship between high glucose (>150 mg/dL) and two modifiable risk factors: body weight and HbA1c. Logistic regression showed that each 1 kg increase in weight raised the odds of high glucose by 2.4% (OR = 1.024), and each 1% rise in HbA1c increased odds over twelve-fold (OR = 12.37), with no statistically significant race-by-predictor interactions detected. With Pacific Islanders as the reference, Black participants had lower adjusted odds (OR = 0.306), while the White and Asian comparisons were not statistically significant.

Observed glucose levels differed among groups; the analysis did not detect statistically significant race-specific differences in the weight and HbA1c associations.

2. Introduction

Effective management of type 2 diabetes (T2DM) depends on early detection of poor blood sugar control and timely treatment. Many studies have shown racial differences in diabetes rates and complications. However, less is known about how baseline fasting glucose levels and their main predictors—body weight and long-term glucose control (HbA1c)—differ across racial groups at first clinic visit. This information is important for setting appropriate screening levels and giving culturally relevant advice.

In this project, we study data from 487 adult male T2DM patients seen at a diabetes clinic in a large city hospital. The patients come from four main racial groups: White, Black, Pacific Islander, and Asian. We focus on two goals:For question one we aim to check if average fasting glucose levels differ by race. If they do, compare each group and test if the average for Whites and Asians differs from that of Blacks and Pacific Islanders.For the second question we want to,

predict the chance of having high blood glucose (>150 mg/dL) based on weight and HbA1c. Also, test if the baseline risk or the effect of these predictors changes by race.

By solving these problems, we want to see how blood glucose levels differ between groups, and whether weight and HbA1c affect them in the same way for all races.

3. Methods

Producing descriptive statistics (sample means, sample standard deviations, minimum, maximum and sample sizes) for fasting blood-glucose (mg/dL) overall and stratified by race/ethnicity (White, Black, Pacific Islander, Asian) is the first step.

To test whether mean blood-glucose levels differed among the four racial/ethnic groups, a classical one-way fixed-effects ANOVA model with restriction is fitted

$$Glucose_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \qquad i = 1, \dots, 4, \qquad j = 1, 2, \dots, n_i,$$

with the constriction that

$$\sum_{\{i=1\}}^{k} a_i = 0, \qquad k = 1, \dots, 4$$

where μ represents an overall mean, α_i is the effect of the i-th race/ethnicity and ε_{ij} ~ Normal(0, σ^2) and k is the number of race/ethnic groups. The constriction is added to solve the overparameterization problem. Since the race/ethnicity group comparisons of mean glucose includes both pairwise, non-pairwise and posterior patterns, Table 1.3 reports contrast t tests; separate Scheffé-adjusted p-values are not provided in the output.

Model diagnostic checks are the third step. We check key assumptions, including independence, homoscedasticity, and normality. To assess homoscedasticity, we compare the sample variances across groups. For normality, we examine the distribution of residuals and use the Shapiro–Wilk test. We also check for influential data points. This includes looking at DFFITs, DFBETAs, and Cook's distance (d_i) to see if any points may strongly affect the model.

If the absolute value of DFFITs is larger than 1 (for 'small' dataset), or $2\sqrt{\frac{p}{n}}$ (for 'large' dataset), where p is number of predictor variables, n is the number of subjects in this analysis,

then the data deserves a closer examination. If the absolute value of DFBETAs is larger than 1 (for 'small' dataset), or $\frac{2}{\sqrt{n}}$ (for 'large' dataset), where n is the number of subjects in this analysis, then the data deserves a closer examination. If the d_i is larger than 1, then this data needs a closer examination.

For question 2, we used a cross sectional dataset of N = 487 adult male type-2 diabetics, imported from an Excel file into SAS, key variables were: race (numeric: 1 = Caucasian, 2 = Black, 3 = Pacific Islander, 4 = Asian), wtkg (weight in kg), gluc (random blood glucose in mg/dL), hba1c (%).

We created the binary outcome as:

$$high_gluc = \begin{cases} 1, & if\ gluc > 150, \\ 0, & otherwise. \end{cases}$$

and confirmed its overall prevalence.

Next, to examine whether the wtkg high_gluc and hba1c high_gluc associations differ by race, we fitted three nested logistic regression models (3 = Pacific Islander serves as the reference category):

Model A (Baseline)

$$P(high_gluc=1) = \beta_0 + \beta_1 \cdot wtkg + \beta_2 \cdot hba1c$$

This quantifies the pooled effects of weight and HbA1c on high glucose.

Model B (Equal Intercept Test)

$$P(high_g luc = 1) = \beta 0 + \beta 1 * wtkg + \beta 2 * hba1c + \sum k = 2..4\, \gamma_k * I(race = k)$$

where I(race=k) is an indicator for race =k. A likelihood-ratio test (LRT) comparing Model A vs. B evaluates whether the race-specific intercepts differ.

Model C (Parallelism Test)

$$logit\left[P\left(high_g luc = 1\right)\right] = \beta 0 + \beta 1 * wtkg + \beta 2 * hba1c + \sum_{k=2..4} \gamma_k * I(race = k) + \sum_{k=2..4} \delta_{1k} * wtkg * I(race = k) + \sum_{k=2..4} \delta_{2k} * hba1c * I(race = k)$$

Here the interaction terms δ_{1k} and δ_{2k} test whether the slopes for wtkg and hba1c vary by race. Finally, we conducted influence diagnostics (Cook's D, DFBETAS) and leverage plots within each race stratum to identify any overly influential observations.

4. Results

4.1 **Results of question 1**

According to Table 1.1, fasting blood-glucose was recorded for 487 samples. For each of these four races group (Caucasian, Black, Pacific Islander and Asian), there is different number of samples (120, 135, 107, 125, respectively, and the overall one is 487), sample mean (143.08, 150.58, 150.77, 139.64, respectively, and the overall one is 145.97) and sample standard deviations (21.57, 24.54, 23.17, 22.69, respectively, and the overall one is 23.49).

According to Table 1.2, a classical one-way fixed-effects ANOVA, fitted via a regression model constrained so that the four race effects summed to zero, produced F(3, 483) = 7.11 with $p = 0.0001$.

According to Table 1.3, the reported post-hoc contrast t tests show the estimated differences in mean blood glucose levels across racial groups. The estimated differences in mean glucose levels were –7.49 mg/dL (standard error (SE) = 2.89, $t = -2.59$, $p = 0.0099$) for White vs Black, –7.68 mg/dL (SE = 3.07, $t = -2.51$, $p = 0.0125$) for White vs Pacific Islander, and 3.44 mg/dL (SE = 2.95, $t = 1.17$, $p = 0.2432$) for White vs Asian. The difference was –0.19 mg/dL (SE = 2.98, $t = -0.06$, $p = 0.9496$) for Black vs Pacific Islander, 10.94 mg/dL (SE = 2.86, $t = 3.82$, $p = 0.0001$) for Black vs Asian, and 11.13 mg/dL (SE = 3.04, $t = 3.66$, $p = 0.0003$) for Pacific Islander vs Asian. The reported combined-group contrast was –18.62 mg/dL (SE = 4.19, t = –4.44, p < 0.0001), corresponding to a difference between the two group averages of –9.31 mg/dL (SE = 2.10).

According to Tables 1.4, 1.7, 1.10 and 1.13, the sample variances of residuals were 465.12 for Caucasian (White) patients, 602.26 for Black patients, 536.84 for Pacific Islander patients, and 514.81 for Asian patients. According to Figures 1.5, 1.8, 1.11, 1.14, the residuals exhibit a distribution that aligns closely with the normal curve for Caucasian (White), Black, Pacific Islander and Asian patients, respectively. According to the Tables 1.6, 1.9, 1.12 and 1.15, the Shapiro-Wilk test yielded W = 0.9919 (p = 0.7161) for White patients, 0.9930 (p = 0.7433) for Black patients, 0.9875 (p = 0.4234) for Pacific Islander patients, and 0.9887 (p = 0.3936) for Asian patients.

According to Tables 1.16, 1.17, 1.18 and 1.19, there are 4, 3, 3, and 4 observations in the White, Black, Pacific Islander, and Asian groups, respectively, with slightly higher DFBETAS and DFFITS values. Based on the large-sample threshold formula, the corresponding cutoff values for DFBETAS and DFFITS are 0.183 for White, 0.172 for Black, 0.193 for Pacific Islander, and 0.179 for Asian groups.

4.2 Results of Question 2

4.2.1 **Descriptive Statistics**

Among the N=487 study participants, 204 (41.89%) had random blood glucose > 150 mg/dL and were thus classified as high_gluc = 1 (Appendix Table 2.1).

4.2.2 **Model A (Baseline)**

For weight (wtkg): OR = 1.024 (95% CI 1.010–1.039), p < 0.001. Each additional kilogram was associated with a 2.4% increase in the odds of high blood glucose. For HbA1c: OR = 12.372 (95% CI 7.599–20.143), p<0.001. Each one-percentage-point increase in HbA_1c was associated with approximately 12.37 times the odds of high blood glucose (Appendix Table 2.2).

4.2.3 **Model B (Equal Intercept)**

Model B included weight, HbA_1c, and race indicators, with Pacific Islanders serving as the reference group (Appendix Table 2.3).

For Black patients, OR = 0.306 (95% CI 0.154–0.610), p=0.0008. After adjusting for weight and HbA_1c, Black participants had 69.4% lower odds of high glucose compared to Pacific Islanders. For Caucasian patients, OR = 0.945 (95% CI 0.481–1.859), p=0.8706. The Caucasian-versus-Pacific Islander comparison was not statistically significant.

For Asian patients, OR = 0.926 (95% CI 0.455–1.885), p = 0.8331. The Asian-versus-Pacific Islander comparison was not statistically significant (Appendix Table 2.3).

4.2.4 **Model C (Parallelism)**

The joint Wald tests for wtkg×race and hba1c×race were not statistically significant ($\chi^2(3) = 0.8764$, $p = 0.8311$; $\chi^2(3) = 2.7154$, $p = 0.4376$, respectively; Appendix Table 2.4).

4.2.5 **Diagnostics**

We assessed influence via the set of diagnostic plots—Pearson and deviance residuals, leverage, Cook's D, C-displacement and deletion differences (Appendix 2.5–2.7). No observation exceeded common thresholds (e.g. no leverage > 0.06, no Cook's D > 0.5, all DFBETAS < 1). However, Case 47 was the most extreme point in several plots but still well within acceptable ranges.

5. Discussion

According to the Table 1.2, for the overall F test, since the p-value (0.0001) is less than 0.05, the null hypothesis, the mean of fasting blood-glucose levels across these four race groups is the same, is rejected at 5% significance level, indicating that fasting blood-glucose levels statistically significantly differed by race/ethnicity at 5% significance level.

According to Table 1.3, Asians had lower observed mean glucose than both Blacks (–10.9 mg/dL, $p = 0.0001$) and Pacific Islanders (–11.1 mg/dL, $p = 0.0003$), and Whites had lower observed mean glucose than both Blacks (–7.49 mg/dL, $p = 0.0099$) and Pacific Islanders (–7.68 mg/dL, $p = 0.0125$) in the reported unadjusted tests. No statistically significant difference emerged between Whites and Asians ($p = 0.2432$) or between Blacks and Pacific Islanders ($p = 0.9496$) since the p-value for these two comparisons are higher than 0.05. The corresponding unadjusted contrast between the two group averages was –9.31 mg/dL ($p < 0.0001$).

The analysis assumes independent observations; independence cannot be established from the reported summary output. For the homoscedasticity, according to Tables 1.4, 1.7, 1.10 and 1.13, except the sample variances of Black patients (602.26) being slightly higher, which is acceptable, the sample variances for the other three groups were broadly similar; these comparisons alone do not establish homoscedasticity.

According to Figures 1.5, 1.8, 1.11, 1.14, the residuals distribution aligns closely with the normal curve for Caucasian (White), Black, Pacific Islander and Asian patients, respectively, which is consistent with the Shapiro–Wilk test results in Tables 1.6, 1.9, 1.12, and 1.15, which yielded $W = 0.9919$ ($p = 0.7161$) for White, 0.9930 ($p = 0.7433$) for Black, 0.9875 ($p = 0.4234$) for Pacific Islander, and 0.9887 ($p = 0.3936$) for Asian. All p-values exceed 0.05, so the Shapiro–Wilk tests did not reject normality within these groups at the 5% significance level.

According to Tables 1.16 through 1.19, there are 4, 3, 3, and 4 observations with slightly elevated DFBETAS and DFFITS values in the White, Black, Pacific Islander, and Asian groups, respectively. These observations exceeded the stated flagging thresholds and warrant further inspection; such flags do not by themselves invalidate the fitted models.

Higher glucose levels in Black and Pacific Islander patients may be related to genetics, diet, income, or differences in access to diabetes care. Lower glucose levels in Asian patients may be due to stricter diets or earlier clinic visits. These results suggest that early care for Black and Pacific Islander men should include culturally appropriate support to reduce early health risks.

One limitation is that this study is cross-sectional and only includes male patients from one urban hospital. These results may not apply to women or people in rural areas. Future research should include women and patients from other settings to improve generalizability.

As for question 2, in this sample of adults with type-2 diabetes, higher weight and HbA_1c were both strong, independent predictors of "high" random glucose: each kilogram increase raised odds by 2.4%, and each 1% rise in HbA_1c increased odds by over twelvefold (Model A). In Model B, Black participants had lower adjusted odds than Pacific Islanders (OR = 0.306, p = 0.0008); White and Asian comparisons with Pacific Islanders were not statistically significant (Appendix Table 2.3).

Weight×race and HbA_1c×race interactions were all non-significant (in Model C), providing no evidence of race-specific differences in these associations in this sample. Given the cross-sectional design, these associations do not establish causal effects or justify changes in screening thresholds. Future studies that follow patients over time and include more information about income and lifestyle may improve these models.

6. References

1. Selvin E, Steffes MW, Ballantyne CM, et al.Racial Differences in Glycemic Markers: A Cross-sectional Analysis of Community-Based Data.*Diabetes Care*. 2009;32(5):787–792.
2. Ziemer DC, Kolm P, Weintraub WS, et al Glucose-independent racial disparity in HbA1c is evident at onset of type 1 diabetes.*Journal of Clinical & Translational Endocrinology*. 2022;27:100284.
3. American Diabetes Association.Racial and Ethnic Disparities in Diabetes Prevalence, Self-Management, and Health Outcomes.*Diabetes Care*. 2017;40(9):e127–e128.

7. Appendix

Table 1.1: Descriptive statistics of blood glucose levels (mg/dL) stratified by race/ethnicity.

	gluc					
	N	Mean	Std	Var	Max	Min
race						
1	120	143.08	21.57	465.12	199.00	92.00
2	135	150.58	24.54	602.26	215.00	95.00
3	107	150.77	23.17	536.84	205.00	93.00
4	125	139.64	22.69	514.81	207.00	86.00
All	487	145.97	23.49	551.71	215.00	86.00

Table 1.2: The one-way fixed-effect ANOVA table with restriction

REG for the classical one-way ANOVA model
(using a restriction to make all alphai effects sum to 0)
(estimates become equivalent to those for effects coding)

The GLM Procedure

Class Level Information		
Class	Levels	Values
race	4	1 2 3 4

Number of Observations Read	487
Number of Observations Used	487

REG for the classical one-way ANOVA model
(using a restriction to make all alphai effects sum to 0)
(estimates become equivalent to those for effects coding)

The GLM Procedure

Dependent Variable: gluc

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	11336.3477	3778.7826	7.11	0.0001
Error	483	256794.0589	531.6647		
Corrected Total	486	268130.4066			

R-Square	Coeff Var	Root MSE	gluc Mean
0.042279	15.79683	23.05786	145.9651

Source	DF	Type I SS	Mean Square	F Value	Pr > F
race	3	11336.34769	3778.78256	7.11	0.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
race	3	11336.34769	3778.78256	7.11	0.0001

Table 1.3: Scheffé-adjusted pairwise and contrast-based comparisons of mean fasting glucose (mg/dL) across race/ethnicity groups

REG for the classical one-way ANOVA model
(using a restriction to make all alphai effects sum to 0)
(estimates become equivalent to those for effects coding)

The GLM Procedure

Dependent Variable: gluc

Parameter	Estimate	Standard Error	t Value	Pr > |t|
White vs Black	-7.4944444	2.89288722	-2.59	0.0099
White vs Pacific Isl	-7.6830218	3.06583895	-2.51	0.0125
White vs Asian	3.4433333	2.94683849	1.17	0.2432
Black vs Pacific Isl	-0.1885774	2.98447421	-0.06	0.9496
Black vs Asian	10.9377778	2.86209275	3.82	0.0001
Pacific Isl vs Asian	11.1263551	3.03679879	3.66	0.0003
White+Asian vs Black+PI	-18.6207996	4.19415586	-4.44	<.0001

Table 1.4: Model summary and ANOVA results for patients in White group.

Regression for race = 1

The REG Procedure
Model: MODEL1
Dependent Variable: gluc

Number of Observations Read	120
Number of Observations Used	120

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	0	0	.	.	.
Error	119	55349	465.11905		
Corrected Total	119	55349			

Root MSE	21.56662	R-Square	0.0000
Dependent Mean	143.08333	Adj R-Sq	0.0000
Coeff Var	15.07277		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	143.08333	1.96875	72.68	<.0001	0

Figure 1.5: Distribution of residuals for the regression model in the White group

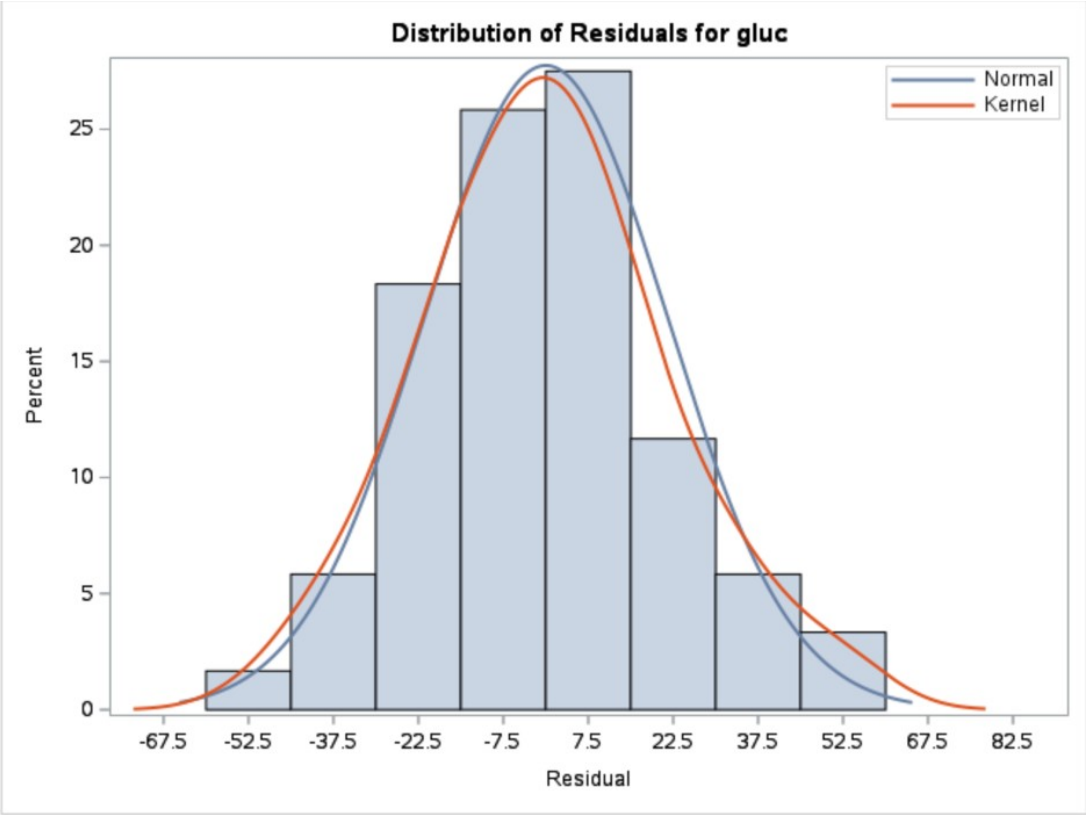


Table 1.6: Shapiro-Wilk and other normality test results for residuals in the White group

Normality Test for Residuals (ONE) for race = 1

The UNIVARIATE Procedure
Variable: Residual

Moments			
N	120	Sum Weights	120
Mean	0	Sum Observations	0
Std Deviation	21.5666188	Variance	465.119048
Skewness	0.21092243	Kurtosis	0.01284802
Uncorrected SS	55349.1667	Corrected SS	55349.1667
Coeff Variation	.	Std Error Mean	1.96875394

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	21.56662
Median	-0.58333	Variance	465.11905
Mode	-4.08333	Range	107.00000
		Interquartile Range	28.50000

Note: The mode displayed is the smallest of 3 modes with a count of 5.

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	-2	Pr >= |M|	0.7843
Signed Rank	S	-119	Pr >= |S|	0.7567

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.991947	Pr < W	0.7161
Kolmogorov-Smirnov	D	0.047824	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.040021	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.260824	Pr > A-Sq	>0.2500

Table 1.7: Model summary and ANOVA results for patients in Black group.

Regression for race = 2

The REG Procedure
Model: MODEL1
Dependent Variable: gluc

Number of Observations Read	135
Number of Observations Used	135

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	0	0	.	.	.
Error	134	80703	602.26070		
Corrected Total	134	80703			

Root MSE	24.54100	R-Square	0.0000
Dependent Mean	150.57778	Adj R-Sq	0.0000
Coeff Var	16.29789		

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	150.57778	2.11215	71.29	<.0001	0

Figure 1.8: Distribution of residuals for the regression model in the Black group

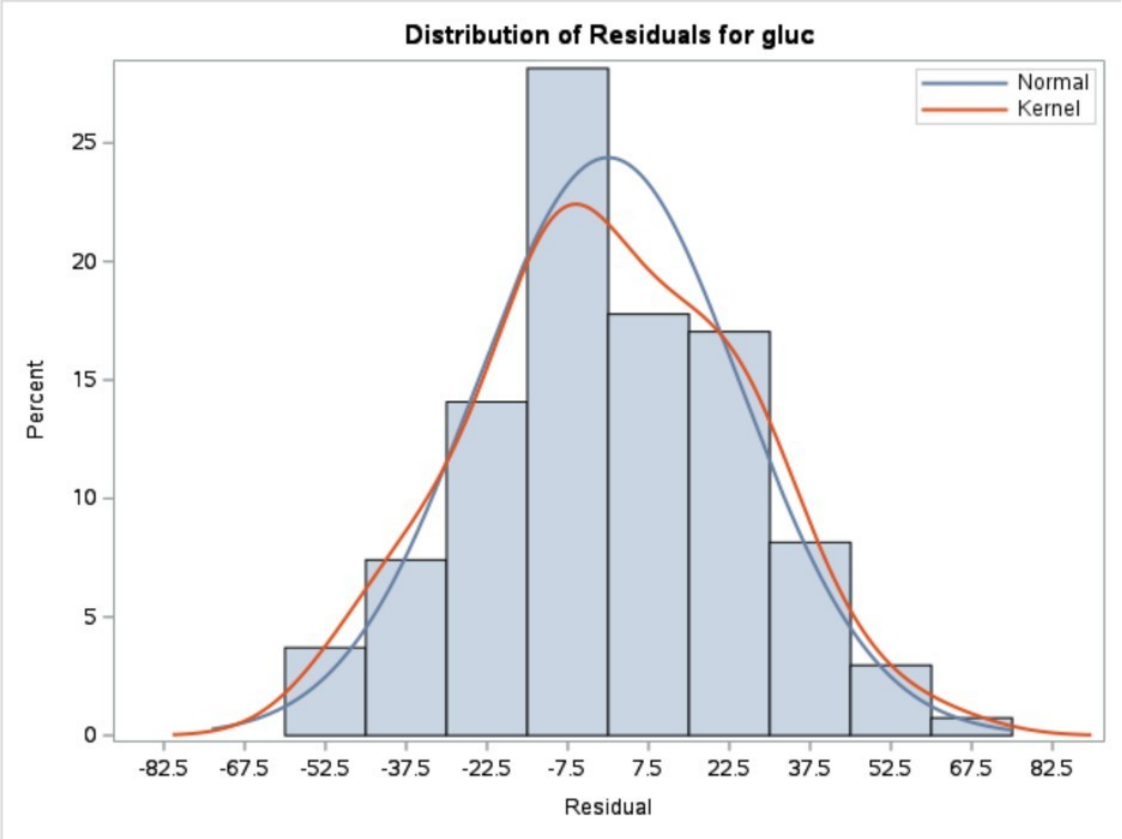


Table 1.9: Shapiro-Wilk and other normality test results for residuals in the Black group

Normality Test for Residuals (ONE) for race = 2

The UNIVARIATE Procedure
Variable: Residual

Moments			
N	135	Sum Weights	135
Mean	0	Sum Observations	0
Std Deviation	24.5410003	Variance	602.260697
Skewness	0.04229567	Kurtosis	-0.3501457
Uncorrected SS	80702.9333	Corrected SS	80702.9333
Coeff Variation	.	Std Error Mean	2.11215301

Basic Statistical Measures			
Location		Variability	
Mean	0.0000	Std Deviation	24.54100
Median	-1.5778	Variance	602.26070
Mode	-13.5778	Range	120.00000
		Interquartile Range	34.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	-4.5	Pr >= |M|	0.4913
Signed Rank	S	-55	Pr >= |S|	0.9044

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.992962	Pr < W	0.7433
Kolmogorov-Smirnov	D	0.049158	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.048562	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.281024	Pr > A-Sq	>0.2500

Table 1.10: Model summary and ANOVA results for patients in the Pacific Islander group

Regression for race = 3

The REG Procedure
Model: MODEL1
Dependent Variable: gluc

Number of Observations Read	107
Number of Observations Used	107

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	0	0	.	.	.
Error	106	56905	536.84112		
Corrected Total	106	56905			

Root MSE	23.16983	R-Square	0.0000
Dependent Mean	150.76636	Adj R-Sq	0.0000
Coeff Var	15.36804		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	150.76636	2.23991	67.31	<.0001	0

Figure 1.11: Distribution of residuals for the regression model in the Pacific Islander group

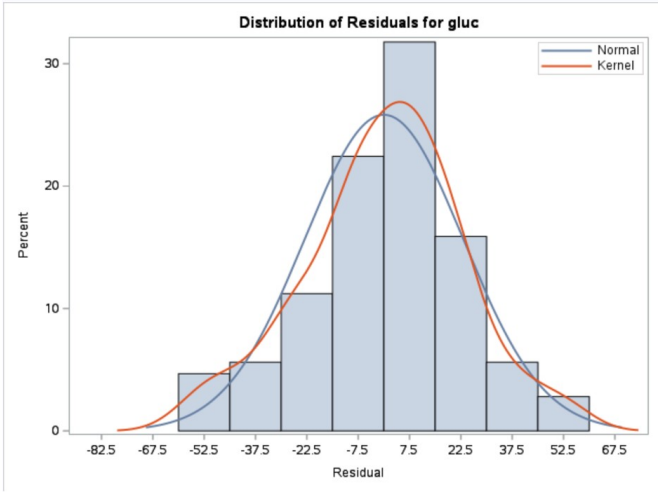


Table 1.12: Shapiro-Wilk and other normality test results for residuals in the Pacific Islander group

Normality Test for Residuals (ONE) for race = 3

The UNIVARIATE Procedure
Variable: Residual

Moments			
N	107	Sum Weights	107
Mean	0	Sum Observations	0
Std Deviation	23.1698321	Variance	536.841121
Skewness	-0.1973622	Kurtosis	0.09069702
Uncorrected SS	56905.1589	Corrected SS	56905.1589
Coeff Variation	.	Std Error Mean	2.23991222

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	23.16983
Median	2.23364	Variance	536.84112
Mode	13.23364	Range	112.00000
		Interquartile Range	26.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	6.5	Pr >= |M|	0.2459
Signed Rank	S	88	Pr >= |S|	0.7859

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.987517	Pr < W	0.4234
Kolmogorov-Smirnov	D	0.064771	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.069985	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.430273	Pr > A-Sq	>0.2500

Table 1.13: Model summary and ANOVA results for patients in the Asian group

Regression for race = 4

The REG Procedure
Model: MODEL1
Dependent Variable: gluc

Number of Observations Read	125
Number of Observations Used	125

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	0	0	.	.	.
Error	124	63837	514.81290		
Corrected Total	124	63837			

Root MSE	22.68949	R-Square	0.0000
Dependent Mean	139.64000	Adj R-Sq	0.0000
Coeff Var	16.24856		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	139.64000	2.02941	68.81	<.0001	0

Figure 1.14: Distribution of residuals for the regression model in the Asian group

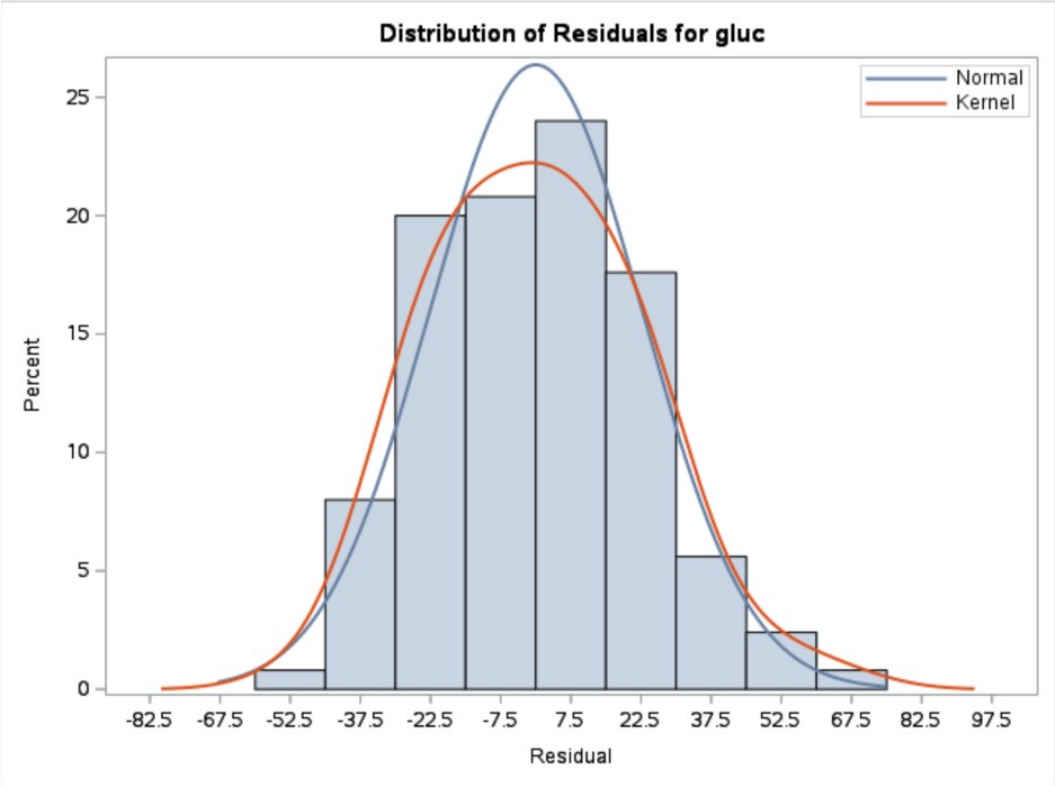


Table 1.15: Shapiro-Wilk and other normality test results for residuals in the Asian group

Normality Test for Residuals (ONE) for race = 4

The UNIVARIATE Procedure
Variable: Residual

Moments			
N	125	Sum Weights	125
Mean	0	Sum Observations	0
Std Deviation	22.6894888	Variance	514.812903
Skewness	0.28996646	Kurtosis	-0.1180465
Uncorrected SS	63836.8	Corrected SS	63836.8
Coeff Variation	.	Std Error Mean	2.02940958

Basic Statistical Measures			
Location		Variability	
Mean	0.000000	Std Deviation	22.68949
Median	0.360000	Variance	514.81290
Mode	4.360000	Range	121.00000
		Interquartile Range	35.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	0.5	Pr >= |M|	1.0000
Signed Rank	S	-124.5	Pr >= |S|	0.7604

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.988678	Pr < W	0.3936
Kolmogorov-Smirnov	D	0.055229	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.056469	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.39155	Pr > A-Sq	>0.2500

Table 1.16: Observations with DFFITS and DFBETAS exceeding the corresponding large-sample threshold in the White patients.

Observations with DFFITS > 0.183 for Race=1

Obs	Observation	DFFITS
46	46	0.2172
51	51	0.2309
102	102	0.2081
109	109	0.2447

Observations with DEBETAS > 0.183 for Race=1

Obs	Observation	DFB_Intercept
46	46	0.2172
51	51	0.2309
102	102	0.2081
109	109	0.2447

Table 1.17: Observations with DFFITS and DFBETAS exceeding the corresponding large-sample threshold in the Black patients.

Observations with DFFITS > 0.172 for Race=2

Obs	Observation	DFFITS
50	50	0.2329
73	73	0.1730
100	100	0.2102

Observations with DEBETAS > 0.172 for Race=2

Obs	Observation	DFB_Intercept
50	50	0.2329
73	73	0.1730
100	100	0.2102

Table 1.18: Observations with DFFITS and DFBETAS exceeding the corresponding large-sample threshold in the Pacific Islander patients.

Observations with DFFITS > 0.193 for Race=3

Obs	Observation	DFFITS
56	56	0.2290
77	77	0.2335
102	102	0.2154

Observations with DEBETAS > 0.193 for Race=3

Obs	Observation	DFB_Intercept
56	56	0.2290
77	77	0.2335
102	102	0.2154

Table 1.19: Observations with DFFITS and DFBETAS exceeding the corresponding large-sample threshold in the Asian patients.

Observations with DFFITS > 0.179 for Race=4

Obs	Observation	DFFITS
18	18	0.2767
84	84	0.2332
95	95	0.2119
119	119	0.1950

Observations with DEBETAS > 0.179 for Race=4

Obs	Observation	DFB_Intercept
18	18	0.2767
84	84	0.2332
95	95	0.2119
119	119	0.1950

Table 2.1

Number of Observations Read	490
Number of Observations Used	487

Response Profile		
Ordered Value	high_gluc	Total Frequency
1	0	283
2	1	204

Probability modeled is high_gluc=1.

Table 2.2

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-21.5124	1.9792	118.1391	<.0001
wtkg	1	0.0242	0.00736	10.8073	0.0010
hba1c	1	2.5154	0.2487	102.3145	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
wtkg	1.024	1.010	1.039
hba1c	12.372	7.599	20.143

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	86.5	Somers' D	0.730
Percent Discordant	13.5	Gamma	0.730
Percent Tied	0.0	Tau-a	0.356
Pairs	57732	c	0.865

Table 2.3

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-24.0832	2.3216	107.6081	<.0001
wtkg		1	0.0304	0.00780	15.2404	<.0001
hba1c		1	2.8314	0.2819	100.8532	<.0001
race	Asian	1	-0.0764	0.3624	0.0444	0.8331
race	Black	1	-1.1841	0.3520	11.3165	0.0008
race	Caucasian	1	-0.0562	0.3449	0.0265	0.8706

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
wtkg	1.031	1.015	1.047
hba1c	16.970	9.765	29.490
race Asian vs Pacific Islander	0.926	0.455	1.885
race Black vs Pacific Islander	0.306	0.154	0.610
race Caucasian vs Pacific Islander	0.945	0.481	1.859

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	87.9	Somers' D	0.759
Percent Discordant	12.1	Gamma	0.759
Percent Tied	0.0	Tau-a	0.370
Pairs	57732	c	0.879

Table 2.4

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	664.253	448.705
SC	668.442	498.964
-2 Log L	662.253	424.705

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	237.5482	11	<.0001
Score	185.8991	11	<.0001
Wald	114.0905	11	<.0001

Joint Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
wtkg	1	3.6865	0.0549
hba1c	1	24.7005	<.0001
race	3	2.1992	0.5321
wtkg*race	3	0.8764	0.8311
hba1c*race	3	2.7154	0.4376

Table 2.5

43	0	0	1.0000	7.2000	46.5000	-0.3108	-0.4294	0.0118	-0.0165	0.00383	-0.00635	-0.0129	0.00998	0.0260	0.00116	0.00115	0.1855	0.0977
44	0	1.0000	0	7.4000	67.7000	-0.3240	-0.4469	0.00931	-0.0201	0.00410	-0.0203	0.00192	0.0163	0.0178	0.000997	0.000987	0.2007	0.1060
45	0	0	1.0000	7.3000	75.1000	-0.5533	-0.7308	0.0111	-0.0183	0.00371	-0.00594	-0.0355	0.0150	0.0156	0.00348	0.00344	0.5376	0.3096
46	0	0	1.0000	7.8000	126.2	0.4091	0.5564	0.0222	-0.0297	0.00705	-0.0117	0.0250	0.0168	0.0507	0.00388	0.00379	0.3133	0.1712
47	1.0000	0	0	10.3000	64.0000	-32.3379	-3.7292	0.000673	0.7249	-0.2957	0.1804	-0.0952	-0.7923	0.0409	0.7046	0.7041	14.6110	1046.4
48	0	0	0	7.8000	82.4000	0.7748	0.9697	0.0152	-0.00445	-0.0638	-0.0720	-0.0677	0.0158	0.000299	0.00939	0.00925	0.9496	0.6095
49	0	0	0	6.6000	93.5000	-0.2795	-0.3879	0.0103	-0.0205	0.0157	0.00876	0.0159	0.0211	-0.00412	0.000822	0.000814	0.1513	0.0789
50	0	0	0	7.6000	83.4000	-0.9873	-1.1666	0.0158	-0.0197	0.0882	0.0884	0.0925	0.00708	-0.00170	0.0159	0.0157	1.3766	0.9905
51	0	0	0	7.8000	66.9000	0.9809	1.1611	0.0198	0.0161	-0.0899	-0.0839	-0.0895	0.0152	-0.0601	0.0198	0.0194	1.3676	0.9816
52	1.0000	0	0	8.0000	86.5000	0.5698	0.7498	0.0172	-0.0423	0.0517	-0.0123	0.00476	0.0400	0.0187	0.00579	0.00569	0.5680	0.3303
53	0	1.0000	0	8.6000	109.2	0.3001	0.4153	0.00852	-0.0193	0.00381	0.00583	0.00196	0.0165	0.0141	0.000781	0.000774	0.1733	0.0908
54	0	0	1.0000	6.7000	63.7000	-0.1989	-0.2786	0.00492	-0.0104	0.00200	-0.00317	-0.00492	0.00921	0.00652	0.000197	0.000196	0.0778	0.0398

Figure 2.6

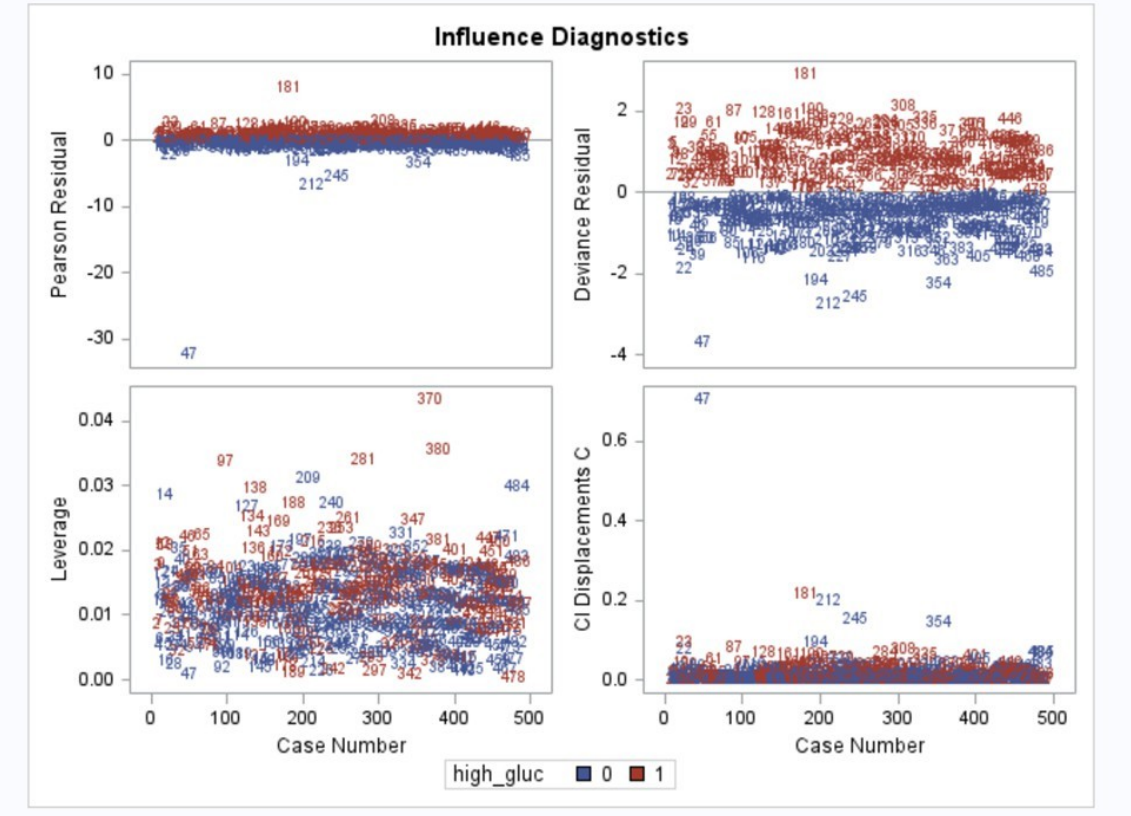
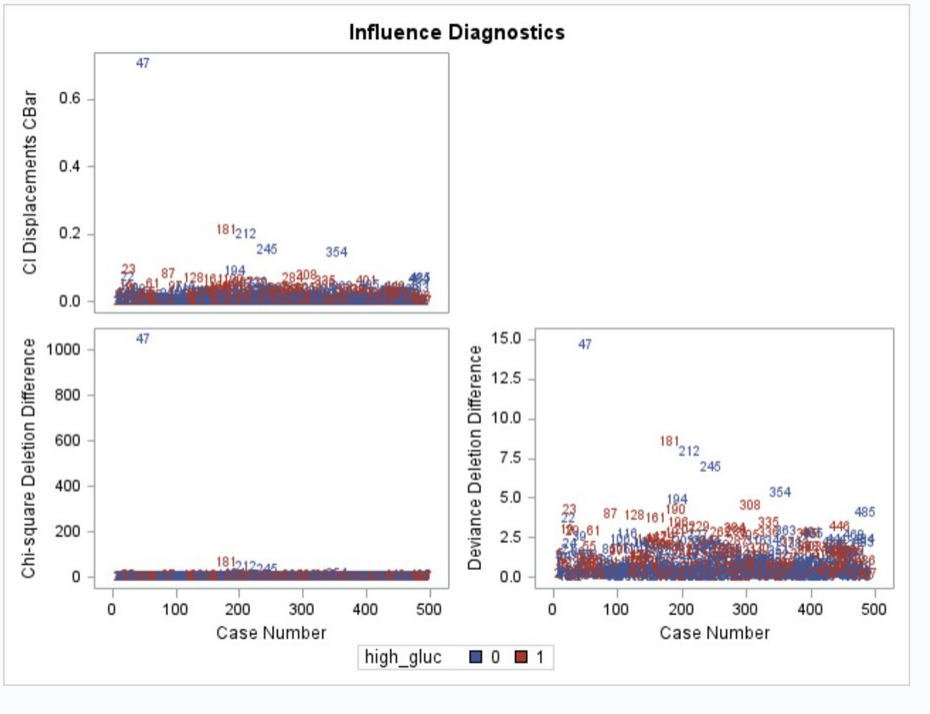


Figure 2.7


Influence Diagnostics
CI Displacements CBar
Chi-square Deletion Difference
Deviance Deletion Difference
Case Number
Case Number
high_gluc 0 1